Math **163** Name (PLEASE PRINT) Cap'n Orangebeard

Exam **2A (Online)** **2020 Fall**/Spring/Summer I /Summer II Score _____

Please fill in the following table with numbers corresponding to parts of problems that you have completed and submitted, totals, and an expected total grade. You will lose double the points that a part is worth if you submit incomplete, unproved, or unchecked answers, as per the directions of the problem. Ideally you should know exactly what you have done correctly and what grade you should receive. I will fill in the received grade after I correct the exam. Not all questions have the same number of parts so that some columns may be redundant for particular problems.

										Expected	Received
	Part	Part	Part	Part	Part	Part	Part	Part	Part	Grade	Grade
	(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)	(viii)	(ix)		
Number											
1a	3	3								6	
1b	3	3								6	
1c											
2a	6									6	
2b	6									6	
2c											
3a	2	4								6	
3b	6									6	
3c											
								Total		36	

Instructions:

Your J-number = 43100000

$a := mnm_{k \in 1..8}\{J_k | (k \in 1..8) \wedge (J_k \neq 0)\} =$ 1

$b := mxm(J_k | k \in 1..8) =$ 4

$c := \left\lfloor \left(\frac{b+a}{2}\right) \right. = \left\lfloor \left(\frac{(4+1)}{2}\right) \right. = \left\lfloor \left(\frac{5}{2}\right) \right. = \left\lfloor \left((2)+\left(\frac{1}{2}\right)\right) \right. =$ 2 =

$d :=$ the smallest **odd** number **strictly greater** than c = 3

6 (1a) Expand using the binomial theorem **EXACTLY AS SHOWN IN CLASS.**

3 (i) $(b - ax)^c$

My problem: $\left((4) + (-1)x\right)^{(2)}$

$= \left(\quad + x \right)^{(\ )} \quad (4 + (-1)x)^2$

$= \quad 16$

$+ \quad (-1)8x$

$+ \quad x^2$

1a.i.0

$$\frac{(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^{n-k} b^k}{(4+(-1)n)^2 = \sum_{k=0}^{2} \binom{2}{k} 4^{2-k} ((-1)n)^k} \left\langle \begin{array}{l} n \longleftarrow 2 \\ a \longleftarrow 4 \\ b \longleftarrow ((-1)n) \end{array} \right\rangle$$

1

$$\sum_{k=0}^{2} \binom{2}{k} 4^{2-k} ((-1)n)^k$$

$$= \left| \begin{array}{l} \binom{2}{0} 4^{2-0} ((-1)n)^0 \\ + \binom{2}{1} 4^{2-1} ((-1)n)^1 \\ + \binom{2}{2} 4^{2-2} ((-1)n)^2 \end{array} \right. = \left| \begin{array}{l} 16 \\ + (-1)8n \\ + nn \end{array} \right. = \left| \begin{array}{l} 16 \\ + (-1)8n \\ + n^2 \end{array} \right.$$

2

$$\frac{\binom{n}{0} = 1}{\binom{2}{0} = 1} \left\langle n \longleftarrow 2 \right\rangle$$

3

$$\frac{\binom{n}{1} = n}{\binom{2}{1} = 2} \left\langle n \longleftarrow 2 \right\rangle$$

4

$$\frac{\binom{n}{n} = 1}{\binom{2}{2} = 1} \left\langle n \longleftarrow 2 \right\rangle$$

5

$$\binom{2}{0} 4^{2-0} ((-1)n)^0$$
$$= (1)(16)(1)$$
$$= 16$$

6

$$\binom{2}{1} 4^{2-1} ((-1)n)^1$$
$$= (2)(4^1)(-1)n$$
$$= (-1)8n$$

7

$$\binom{2}{2} 4^{2-2} ((-1)n)^2$$
$$= (1)(4^0)(1)nn$$
$$= nn$$

(1a) Expand using the multinomial theorem **EXACTLY AS SHOWN IN CLASS.**

3 (ii) $(c - x + y)^{1+a}$

My problem: $\left((2) + (-1)x + y\right)^{1+(1)}$

$= \left((2) + (-1)x + y\right)^{(2)}$

$= 4$

$+ (-1)4x$

$+ x^2$

$+ (-1)2xy$

$+ 4y$

$+ y^2$

1a.ii.0

$$(x_1 + x_2 + \cdots + x_p)^n = \sum_{n_1+n_2+\cdots+n_p=n} \binom{n}{n_1\ n_2\ \cdots\ n_p} (x_1^{n_1})(x_2^{n_2}) \cdots (x_p^{n_p})$$

$$(2 + (-1)x + y)^2 = \sum_{n_1+n_2+n_3=2} \binom{2}{n_1\ n_2\ n_3} (2^{n_1})(((-1)x)^{n_2})(y^{n_3})$$

$p \leftarrow 3$
$n \leftarrow 2$
$x_1 \leftarrow 2$
$x_2 \leftarrow (-1)x$
$x_3 \leftarrow y$

1

$$\begin{aligned} 2 &= 0+0+2 = 0+1+1 \\ &= 0+2+0 = 1+0+1 \\ &= 2+0+0 = 1+1+0 \end{aligned}$$

2

$$\sum_{n_1+n_2+n_3=2} \binom{2}{n_1\ n_2\ n_3} (2^{n_1})(((-1)x)^{n_2})(y^{n_3})$$

$$= \binom{2}{0\ 0\ 2} 2^0 ((-1)x)^0 y^2 = yy = 4 = 4$$
$$+ \binom{2}{0\ 1\ 1} 2^0 ((-1)x)^1 y^1 + (-1)2xy + (-1)4x + (-1)4x$$
$$+ \binom{2}{0\ 2\ 0} 2^0 ((-1)x)^2 y^0 + xx + xx + x^2$$
$$+ \binom{2}{1\ 0\ 1} 2^1 ((-1)x)^0 y^1 + 4y + (-1)2xy + (-1)2xy$$
$$+ \binom{2}{2\ 0\ 0} 2^2 ((-1)x)^0 y^0 + 4 + 4y + 4y$$
$$+ \binom{2}{1\ 1\ 0} 2^1 ((-1)x)^1 y^0 + (-1)4x + yy + y^2$$

3

$$\begin{aligned} &\binom{2}{0\ 0\ 2} \\ &= \binom{2}{0\ 2\ 0} \\ &= \binom{2}{2\ 0\ 0} \\ &= \frac{\cancel{(2!)}}{(0!)\cancel{(2!)}(0!)} \\ &= \frac{1}{(1)(1)(1)} \\ &= 1 \end{aligned}$$

4

$$\begin{aligned} &\binom{2}{0\ 1\ 1} \\ &= \binom{2}{1\ 0\ 1} \\ &= \binom{2}{1\ 1\ 0} \\ &= \frac{(2!)}{(1!)(1!)(0!)} \\ &= \frac{2}{(1)(1)(1)} \\ &= 2 \end{aligned}$$

5

$$\begin{aligned} &\binom{2}{0\ 0\ 2} 2^0 ((-1)x)^0 y^2 \\ &= (1)(1)(1)\,yy \\ &= yy \end{aligned}$$

6

$$\begin{aligned} &\binom{2}{0\ 1\ 1} 2^0 ((-1)x)^1 y^1 \\ &= (2)(1)(-1)xy \\ &= (-1)2xy \end{aligned}$$

7

$$\begin{aligned} &\binom{2}{0\ 2\ 0} 2^0 ((-1)x)^2 y^0 \\ &= (1)(1)(1)(1)\,xx \\ &= xx \end{aligned}$$

8

$$\begin{aligned} &\binom{2}{1\ 0\ 1} 2^1 ((-1)x)^0 y^1 \\ &= (2)(2)(1)\,y \\ &= 4y \end{aligned}$$

9

$$\begin{aligned} &\binom{2}{2\ 0\ 0} 2^2 ((-1)x)^0 y^0 \\ &= (1)(4)(1)(1) \\ &= 4 \end{aligned}$$

10

$$\begin{aligned} &\binom{2}{1\ 1\ 0} 2^1 ((-1)x)^1 y^0 \\ &= (2)(2)(-1)x(1) \\ &= (-1)4x \end{aligned}$$

6 (1b) Solve the following recurrence relations with the indicated initial conditions, using the formula for second-order recurrences given in class.

3 (i)
$$\begin{cases} \rho(0) \coloneqq 1 \\ \rho(1) \coloneqq 2 \\ \forall n \in \mathbb{N} \quad \Big(\rho(n+2)\Big) - (b+c)\Big(\rho(n+1)\Big) + (bc)\Big(\rho(n)\Big) = 0 \end{cases}$$

First rewrite the recurrence with your values of ***b*** **and** ***c***.

My problem:

$$\Big(\rho(n+2)\Big) - \Big((4)+(2)\Big)\Big(\rho(n+1)\Big) + \Big((4)(2)\Big)\Big(\rho(n)\Big) = 0$$

My problem:

$$\Big(1\Big)\Big(\rho(n+2)\Big) + \Big(-6\Big)\Big(\rho(n+1)\Big) + (8)\Big(\rho(n)\Big) = 0$$

My solution: $\forall n \in \mathbb{N}$

$$\rho(n)$$

$$= \quad 2^n$$

$$\begin{array}{l} \rho(n+2) - 6\rho(n+1) + 8\rho(n) = 0 \\ \quad -\left(-6\rho(n+1) + 8\rho(n)\right) \quad -\left(-6\rho(n+1) + 8\rho(n)\right) \\ \hline (1)(\rho(n+2)) = -(-6)(\rho(n+1)) - (8)(\rho(n)) \end{array}$$

1

$$\begin{array}{c} (1)n^2 + (-6)n + 8 = 0 \\ \hline (n-4)(n-2) = 0 \\ \hline (n-4=0) \ \vee \ (n-2=0) \\ \hline (n=4) \ \vee \ (n=2) \\ \hline (r_1 = 2) \ \wedge \ (r_2 = 4) \\ \hline r_1 \neq r_2 \\ \hline \forall n \in \mathbb{N} \quad \rho(n) = k_1(2)^n + k_2(4)^n \end{array}$$

$$\begin{aligned} \forall n \in \mathbb{N} \qquad \rho(n) &= (1)(2)^n + (0)(4)^n \\ &= 2^n \end{aligned}$$

2)

$$\frac{\rho(n) = k_1(2)^n + k_2(4)^n \quad \langle n \leftarrow 0 \rangle}{\rho(0) = k_1(2)^0 + k_2(4)^0} \qquad \rho(0) := 1$$

$$1 = k_1(1) + k_2(1)$$

$$\frac{\rho(n) = k_1(2)^n + k_2(4)^n \quad \langle n \leftarrow 1 \rangle}{\rho(1) = k_1(2)^1 + k_2(4)^1} \qquad \rho(1) := 2$$

$$2 = k_1(2) + k_2(4)$$

3)

$$\frac{1 = k_1(1) + k_2(1)}{k_2 = 1 - k_1} \qquad \frac{2 = k_1(2) + k_2(4)}{k_1(2) = 2 - k_2(4)}$$

$$\begin{array}{c} k_1(2) = 2 - (1 - k_1)(4) \\ \hline k_1(2) = 2 - 4 + k_1(4) \\ \hline k_1(2) = -2 + k_1(4) \\ -k_1(2) \qquad -k_1(2) \\ \hline 0 = -2 + k_1(2) \\ \hline k_1(2) = 2 \\ \div 2 \quad \div 2 \\ \hline k_1 = 1 \end{array}$$

$$\begin{aligned} k_2 &= 1 - k_1 \\ &= 1 - 1 \\ &= 0 \end{aligned}$$

(1b) Continued. Solve the following recurrence relations with the indicated initial conditions, using the formula for second-order recurrences given in class, using substitution. Prove by substitution that your answer is correct.

3 (ii)
$$\begin{cases} \rho(0) \coloneqq 1 \\ \rho(1) \coloneqq 2 \\ \forall n \in \mathbb{N} \quad (a^2)\Big(\ \rho(n+2)\Big) - (2ac)\Big(\rho(n+1)\Big) + (c^2)\Big(\rho(n)\Big) = 0 \end{cases}$$

First rewrite with your values of ***a*** **and** ***c***.

My problem:

$$\Big((\ |\ )^2\Big)\Big(\ \rho(n+2)\Big) - \Big(2(\ |\ )(2)\Big)\Big(\rho(n+1)\Big) + \Big((2)^2\Big)\Big(\rho(n)\Big) = 0$$

My problem:

$$\Big(\ |\ \Big)\Big(\ \rho(n+2)\Big) + \Big(-4\quad\Big)\Big(\rho(n+1)\Big) + (4)\Big(\rho(n)\Big) = 0$$

My solution: $\forall n \in \mathbb{N}$

$\rho(n)$

$= \quad 2^n$

16. ii. 0

$$(1)(\rho(n+2)) + (-4)(\rho(n+1)) + (4)(\rho(n)) = 0$$
$$- ((-4)(\rho(n+1)) + (4)(\rho(n))) \quad - ((-4)(\rho(n+1)) + (4)(\rho(n)))$$
$$(1)(\rho(n+2)) = (4)(\rho(n+1)) - (4)(\rho(n))$$

5)

$$\forall n \in \mathbb{N} \left(\begin{array}{c} P(n) \\ := (\rho(n) = 2^{(n)}) \end{array} \right)$$

1)

$$(1)x^2 + (-4)x + 4 = 0$$
$$(x-2)^2 = 0$$
$$x = 2$$
$$r_1 = x = r_2$$
$$\forall n \in \mathbb{N} \quad \rho(n) = k_1((2)^n) + k_2(n(2)^n)$$

$\forall n \in \mathbb{N}$

$$\begin{aligned} \rho(n) &= (1)((2)^n) + (0)(n(2)^n) \\ &= 2^n \end{aligned}$$

6)

$$\frac{P(n) := (\rho(n) = 2^{(n)})}{P(0) := (\rho(0) = 2^{(0)})} \langle n \leftarrow 0 \rangle$$

7)

$$\frac{P(n) := (\rho(n) = 2^{(n)})}{P(1) := (\rho(1) = 2^{(1)})} \langle n \leftarrow 1 \rangle$$

8)

BC ⓪

$$\rho(0) := 1 \quad 2^0 = 1$$
$$\rho(0) = 1 = 2^0$$
$$\rho(0) = 2^0$$
$$P(0)$$

9)

BC ①

$$\rho(1) := 2 \quad 2^1 = 2$$
$$\rho(1) = 2 = 2^1$$
$$\rho(1) = 2^1$$
$$P(1)$$

2)

$$\rho(n) = k_1((2)^n) + k_2(n(2)^n) \quad \langle n \leftarrow 0 \rangle$$
$$\begin{aligned} \rho(0) &= k_1((2)^0) + k_2((0)(2)^0) \\ &= k_1(1) + (0) \\ &= k_1 \end{aligned} \qquad \rho(0) := 1$$
$$k_1 = 1$$

10)

$$\frac{P(n) := (\rho(n) = 2^{(n)})}{P((n+2)) := (\rho((n+2)) = 2^{(n+2)})} \langle n \leftarrow (n+2) \rangle$$

3)

$$\rho(n) = k_1((2)^n) + k_2(n(2)^n) \quad \langle n \leftarrow 1 \rangle$$
$$\begin{aligned} \rho(1) &= k_1((2)^1) + k_2((1)(2)^1) \\ &= k_1(2) + k_2(2) \end{aligned} \qquad \rho(1) := 2$$
$$2 = k_1(2) + k_2(2)$$
$$\div 2 \qquad \div 2$$
$$1 = k_1 + k_2$$

11)

RCS $\quad P(0) \; P(1) \; \ldots \; P(n) \; P(n+1)$

$$\begin{aligned} &\rho(n+2) \\ &= (4)(\rho(n+1)) - (4)(\rho(n)) \\ &= (4)(\rho(n+1) - \rho(n)) \\ &= (4)(2^{(n+1)} - 2^n) \\ &= (4)(2^n)((2)-(1)) \\ &= (4)(2^n)(1) \\ &= (2^2)(2^n)(1) \\ &= 2^{(n+2)} \end{aligned}$$
$$\rho(n+2) = 2^{(n+2)}$$
$$P(n+2)$$

4)

$$k_1 = 1$$
$$k_1 + k_2 = 1$$
$$1 + k_2 = 1$$
$$-1 \qquad -1$$
$$k_2 = 0$$

12)

$$\frac{P(0) \quad \left(\frac{P(0) \; P(1) \; \ldots \; P(n)}{P(n+1)} \right)}{\forall n \in \mathbb{N} \, (P(n))}$$
$$\forall n \in \mathbb{N} \, (\rho(n) = 2^n)$$

4 | 1 | 0

6 (2a) Consider the following recurrence relation for $\varphi(n)$ and $\alpha(n)$ and **answer the questions**:

$$\begin{cases} & \alpha(0) \coloneqq a \\ \forall n \in \mathbb{N} & \alpha(n+1) \coloneqq \big(\varphi(n)\big)b \end{cases}$$

$$\begin{cases} & \varphi(0) \coloneqq \alpha(0) + c \\ \forall n \in \mathbb{N} & \varphi(n+1) \coloneqq \big(\alpha(n)\big)d + a \end{cases}$$

Show calculations on the facing side.

Rewrite the recursion using your values of b, c, and d.

My problem:

$$\begin{cases} & \alpha(0) \coloneqq (\,1\,) \\ \forall n \in \mathbb{N} & \alpha(n+1) \coloneqq \big(\varphi(n)\big)(4) \end{cases} \qquad \begin{cases} & \varphi(0) \coloneqq \alpha(0) + c = (\,1\,) + (2) \\ \forall n \in \mathbb{N} & \varphi(n+1) \coloneqq \big(\alpha(n)\big)(3) + (\,1\,) \end{cases}$$

Show full computations on the facing side for all values and your conjecture:

	$\alpha(0)$		$\varphi(0)$
=	1	=	3
	$\alpha(1)$		$\varphi(1)$
=	12	=	4
	$\alpha(2)$		$\varphi(2)$
=	16	=	37
	$\alpha(3)$		$\varphi(3)$
=	148	=	49

Conjecture : $\forall n \in \mathbb{N}$

	$\alpha(n)$		$\varphi(n)$
=	See following page	=	See following page

2a | 0

$$\begin{array}{l} \alpha(n+1) = (\varphi(n))(4) \\ \hline \alpha(0+1) = (\varphi(0))(4) \\ \hline \alpha(1) = (3)(4) \\ \quad\quad = 12 \end{array} \langle n \leftarrow 0 \rangle$$

1

$$\begin{array}{l} \varphi(n+1) = (\alpha(n))(3)+1 \\ \hline \varphi(0+1) = (\alpha(0))(3)+1 \\ \hline \varphi(1) = (1)(3)+1 \\ \quad\quad = 3+1 \\ \quad\quad = 4 \end{array} \langle n \leftarrow 0 \rangle$$

2

$$\begin{array}{l} \alpha(n+1) = (\varphi(n))(4) \\ \hline \alpha(1+1) = (\varphi(1))(4) \\ \hline \alpha(2) = (4)(4) \\ \quad\quad = 16 \end{array} \langle n \leftarrow 1 \rangle$$

3

$$\begin{array}{l} \varphi(n+1) = (\alpha(n))(3)+1 \\ \hline \varphi(1+1) = (\alpha(1))(3)+1 \\ \hline \varphi(2) = (12)(3)+1 \\ \quad\quad = 36+1 \\ \quad\quad = 37 \end{array} \langle n \leftarrow 1 \rangle$$

4

$$\begin{array}{l} \alpha(n+1) = (\varphi(n))(4) \\ \hline \alpha(2+1) = (\varphi(2))(4) \\ \hline \alpha(3) = (37)(4) \\ \quad\quad = 148 \end{array} \langle n \leftarrow 2 \rangle$$

5

$$\begin{array}{l} \varphi(n+1) = (\alpha(n))(3)+1 \\ \hline \varphi(2+1) = (\alpha(2))(3)+1 \\ \hline \varphi(3) = (16)(3)+1 \\ \quad\quad = 48+1 \\ \quad\quad = 49 \end{array} \langle n \leftarrow 2 \rangle$$

6

$$\begin{aligned} \alpha(n+1) &:= (\varphi(n))(4) \\ &= ((\alpha(n-1))(3)+1))(4) \\ &= (12)(\alpha(n-1))+4 \end{aligned}$$

7

$$\begin{aligned} \varphi(n+1) &:= (\alpha(n))(3)+1 \\ &= ((\varphi(n-1))(4))(3)+1 \\ &= (12)(\varphi(n-1))+1 \end{aligned}$$

8

$$\begin{aligned} & \sum_{k=0}^{\left(\left\lfloor\frac{n}{2}\right\rfloor\right)-1} (12)^k \\ &= S_m((12)^n, \left(\left\lfloor\frac{n}{2}\right\rfloor\right)-1)) \\ &= \frac{1-(12)^{\left(\left(\left\lfloor\frac{n}{2}\right\rfloor\right)-1)+1)}}{1-(12)} \\ &= \frac{(12)^{\left(\left\lfloor\frac{n}{2}\right\rfloor\right)}-1}{11} \end{aligned}$$

9

$$\begin{aligned} \alpha(0) &= 1 \\ \alpha(1) &= 12 \\ \alpha(2) &= (12)(1)+4 \\ \alpha(3) &= (12)(12)+4 \\ \alpha(4) &= (12)((12)(1)+4)+4 = (12)^2(1)+(12)(4)+4 \\ &\vdots \qquad\qquad \vdots \end{aligned}$$

$$\begin{aligned} & \alpha(n) \\ &= (12)^{\left(\left\lfloor\frac{n}{2}\right\rfloor\right)}\left((1)\left(\frac{1+(-1)^n}{2}\right) + (12)\left(\frac{1+(-1)^{(n+1)}}{2}\right)\right) + (4)\left(\sum_{k=0}^{\left(\left(\left\lfloor\frac{n}{2}\right\rfloor\right)-1\right)} (12)^k\right) \\ &= (12)^{\left(\left\lfloor\frac{n}{2}\right\rfloor\right)}\left((1)\left(\frac{1+(-1)^n}{2}\right) + (12)\left(\frac{1+(-1)^{(n+1)}}{2}\right)\right) + (4)\left(\frac{(12)^{\left(\left\lfloor\frac{n}{2}\right\rfloor\right)}-1}{11}\right) \end{aligned}$$

10

$$\begin{aligned} \varphi(0) &= 3 \\ \varphi(1) &= 4 \\ \varphi(2) &= (12)(3)+1 \\ \varphi(3) &= (12)(4)+1 \\ \varphi(4) &= (12)((12)(3)+1)+1 = (12)^2(3)+(12)(1)+1 \\ &\vdots \qquad\qquad \vdots \end{aligned}$$

$$\begin{aligned} & \varphi(n) \\ &= (12)^{\left(\left\lfloor\frac{n}{2}\right\rfloor\right)}\left((3)\left(\frac{1+(-1)^n}{2}\right) + (4)\left(\frac{1+(-1)^{(n+1)}}{2}\right)\right) + (1)\left(\sum_{k=0}^{\left(\left(\left\lfloor\frac{n}{2}\right\rfloor\right)-1\right)} (12)^k\right) \\ &= (12)^{\left(\left\lfloor\frac{n}{2}\right\rfloor\right)}\left((3)\left(\frac{1+(-1)^n}{2}\right) + (4)\left(\frac{1+(-1)^{(n+1)}}{2}\right)\right) + (1)\left(\frac{(12)^{\left(\left\lfloor\frac{n}{2}\right\rfloor\right)}-1}{11}\right) \end{aligned}$$

6 (2b) Find conjectures for the sums:

$$Sm(\alpha, n) \coloneqq S(\alpha, n) \coloneqq \sum_{k=0}^{n} \alpha(k) \quad \text{and}$$

$$Sm(\varphi, n) \coloneqq S(\varphi, n) \coloneqq \sum_{k=0}^{n} \varphi(k)$$

where $\alpha(n)$ and $\varphi(n)$ are defined, as in (2a) by the recursion appearing below,

$$\begin{cases} & \boldsymbol{\alpha(0) \coloneqq a} \\ \forall \boldsymbol{n} \in \mathbb{N} & \boldsymbol{\alpha(n+1) \coloneqq (\varphi(n))b} \end{cases} \qquad \begin{cases} & \boldsymbol{\varphi(0) \coloneqq \alpha(0) + c} \\ \forall \boldsymbol{n} \in \mathbb{N} & \boldsymbol{\varphi(n+1) \coloneqq (\alpha(n))d + a} \end{cases}$$

Show computations on the facing side to indicate how you arrive at your conjecture.

Rewrite the recursion using your values of a, b, and c.

My problem:

$\varphi(0) \coloneqq \alpha(0) + c = (1) + (2) = 3$

$$\begin{cases} & \alpha(0) \coloneqq (\ \vert\ ) \\ \forall n \in \mathbb{N} & \alpha(n+1) \coloneqq (\varphi(n))(4) \end{cases} \qquad \begin{cases} & \cancel{\varphi(0) \coloneqq \alpha(0) = (\ \ )} \\ \forall n \in \mathbb{N} & \varphi(n+1) \coloneqq (\alpha(n))(3) + (\ \vert\ ) \end{cases}$$

My Conjectures :

(1) $\forall n \in \mathbb{N}$ $Sm(\alpha, n)$

$$= \left(\frac{15}{11}\right)\left(\frac{(12)^{\left(\left(\left\lfloor \frac{n}{2} \right\rfloor\right)+1\right)} - 1}{11}\right) + \left(\frac{136}{11}\right)\left(\frac{(12)^{\left(\left(\left\lfloor \frac{(n-1)}{2} \right\rfloor\right)+1\right)} - 1}{11}\right) - \left(\frac{4}{11}\right)(n+1)$$

(2) $\forall n \in \mathbb{N}$ $Sm(\varphi, n)$

$$= \left(\frac{34}{11}\right)\left(\frac{(12)^{\left(\left(\left\lfloor \frac{n}{2} \right\rfloor\right)+1\right)} - 1}{11}\right) + \left(\frac{45}{11}\right)\left(\frac{(12)^{\left(\left(\left\lfloor \frac{(n-1)}{2} \right\rfloor\right)+1\right)} - 1}{11}\right) - \left(\frac{1}{11}\right)(n+1)$$

$$\begin{aligned}
\alpha(0) &= 1 \\
\alpha(1) &= 12 \\
\alpha(2) &= (12)(1) + 4 \\
\alpha(3) &= (12)(12) + 4 \\
\alpha(4) &= (12)\big((12)(1)+4\big) + 4 = (12)^2(1) + (12)(4) + 4
\end{aligned}$$

$$\begin{aligned}
Sm(\alpha, 4) = \quad & (1)(1 + 12 + (12)^2) + (4)((0) + (1) + (1 + 12)) \\
+ \quad & (12)(1 + 12) + (4)((0) + (1)) \\
= \quad & (1)\sum_{k=0}^{2}((12)^k) + (4)\left(\sum_{k=0}^{-1}((12)^k) + \sum_{k=0}^{0}((12)^k) + \sum_{k=0}^{1}((12)^k)\right) \\
+ \quad & (12)\sum_{k=0}^{2-1}((12)^k) + (4)\left(\sum_{k=0}^{-1}((12)^k) + \sum_{k=0}^{0}((12)^k)\right)
\end{aligned}$$

$$\begin{aligned}
& Sm(\alpha, 4) \\
= \quad & \left((1)(Sm(12,2)) + (4)\left(Sm(12,-1) + Sm(12,0) + Sm(12,1)\right)\right) + \left((12)(Sm(12,2-1)) + (4)\left(Sm(12,-1) + Sm(12,0)\right)\right) \\
= \quad & \left((1)(Sm(12,2)) + (4)\left(\frac{(12)^0-1}{12-1} + \frac{(12)^1-1}{12-1} + \frac{(12)^2-1}{12-1}\right)\right) + \left((12)(Sm(12,2-1)) + (4)\left(\frac{(12)^0-1}{12-1} + \frac{(12)^1-1}{12-1}\right)\right) \\
= \quad & \left((1)(Sm(12,2)) + \left(\frac{4}{12-1}\right)\left((12)^0-1) + ((12)^1-1) + ((12)^2-1)\right)\right)\right) + \left((12)(Sm(12,2-1)) + \left(\frac{4}{11}\right)\left((12)^0-1) + ((12)^1-1)\right)\right)\right) \\
= \quad & \left((1)(Sm(12,2)) + \left(\frac{4}{11}\right)\left(((12)^0 + (12)^1 + (12)^2) - (1+1+1)\right)\right) + \left((12)(Sm(12,2-1)) + \left(\frac{4}{11}\right)\left(((12)^0 + (12)^1) - (1+1)\right)\right) \\
= \quad & \left((1)(Sm(12,2)) + \left(\frac{4}{11}\right)\left(\left(\sum_{k=0}^{2}((12)^k)\right) - (1)(2+1)\right)\right) + \left((12)(Sm(12,2-1)) + \left(\frac{4}{11}\right)\left(\left(\sum_{k=0}^{2-1}((12)^k)\right) - (1)(1+1)\right)\right)
\end{aligned}$$

$$\begin{aligned}
& Sm(\alpha, n) \\
= \quad & \left((1)(Sm(12,\lfloor \tfrac{n}{2} \rfloor)) + \left(\frac{4}{11}\right)\left((Sm(12,\lfloor \tfrac{n}{2} \rfloor) - (\lfloor \tfrac{n}{2} \rfloor) - 1)\right)\right) + \left((12)(Sm(12,\lfloor \tfrac{(n-1)}{2} \rfloor)) + \left(\frac{4}{11}\right)\left((Sm(12,\lfloor \tfrac{(n-1)}{2} \rfloor) - (\lfloor \tfrac{(n-1)}{2} \rfloor) - 1)\right)\right) \\
= \quad & \left((1) + \left(\frac{4}{11}\right)\right)(Sm(12,\lfloor \tfrac{n}{2} \rfloor)) + ((12)+(4))(Sm(12,\lfloor \tfrac{(n-1)}{2} \rfloor)) - \left(\frac{4}{11}\right)\left((\lfloor \tfrac{(n-1)}{2} \rfloor) + 1 + (\lfloor \tfrac{n}{2} \rfloor) + 1\right) \\
= \quad & \left(\frac{15}{11}\right)(Sm(12,\lfloor \tfrac{n}{2} \rfloor)) + \left(\frac{136}{11}\right)(Sm(12,\lfloor \tfrac{(n-1)}{2} \rfloor)) - \left(\frac{4}{11}\right)((n-1)+2) \\
= \quad & \left(\frac{15}{11}\right)\left(\frac{(12)^{(\lfloor \frac{n}{2} \rfloor + 1)} - 1}{11}\right) + \left(\frac{136}{11}\right)\left(\frac{(12)^{(\lfloor \frac{(n-1)}{2} \rfloor + 1)} - 1}{11}\right) - \left(\frac{4}{11}\right)(n+1)
\end{aligned}$$

1

$$\begin{aligned}
& \sum_{k=0}^{-1}((12)^k) \\
= \quad & Sm((12)^k, -1) \\
= \quad & \frac{1-(12)^{((-1)+1)}}{1-(12)} \\
= \quad & \frac{(12)^0 - 1}{11} \\
= \quad & \frac{1-1}{11} \\
= \quad & \frac{0}{11} \\
= \quad & 0
\end{aligned}$$

26. ii | 0

$$\forall p \in \mathbb{N}(Evn(p))$$

$$\varphi(n) = (12)^{\left\lfloor \left(\frac{n}{2}\right) \right\rfloor}\left((3)\left(\frac{1+(-1)^n}{2}\right) + (4)\left(\frac{1+(-1)^{(n+1)}}{2}\right)\right) + (1)\left(\frac{(12)^{\left\lfloor \left(\frac{n}{2}\right) \right\rfloor} - 1}{11}\right) \quad \langle n \leftarrow p \rangle$$

$$\varphi(p) = (12)^{\left\lfloor \left(\frac{p}{2}\right) \right\rfloor}\left((3)\left(\frac{1+(-1)^p}{2}\right) + (4)\left(\frac{1+(-1)^{(p+1)}}{2}\right)\right) + (1)\left(\frac{(12)^{\left\lfloor \left(\frac{p}{2}\right) \right\rfloor} - 1}{11}\right)$$

$$= (12)^{\frac{p}{2}}\left((3)(1) + (4)(0)\right) + (1)\left(\frac{(12)^{\frac{p}{2}} - 1}{11}\right)$$

$$= (12)^{\frac{p}{2}}(3) + \left(\frac{1}{11}\right)(12)^{\frac{p}{2}} - \left(\frac{1}{11}\right)$$

1

$$\forall q \in \mathbb{N}(Odd(q))$$

$$\varphi(n) = (12)^{\left\lfloor \left(\frac{n}{2}\right) \right\rfloor}\left((3)\left(\frac{1+(-1)^n}{2}\right) + (4)\left(\frac{1+(-1)^{(n+1)}}{2}\right)\right) + (1)\left(\frac{(12)^{\left\lfloor \left(\frac{n}{2}\right) \right\rfloor} - 1}{11}\right) \quad \langle n \leftarrow q \rangle$$

$$\varphi(q) = (12)^{\left\lfloor \left(\frac{q}{2}\right) \right\rfloor}\left((3)\left(\frac{1+(-1)^q}{2}\right) + (4)\left(\frac{1+(-1)^{(q+1)}}{2}\right)\right) + (1)\left(\frac{(12)^{\left\lfloor \left(\frac{q}{2}\right) \right\rfloor} - 1}{11}\right)$$

$$= (12)^{\frac{(q-1)}{2}}\left((3)(0) + (4)(1)\right) + (1)\left(\frac{(12)^{\frac{(q-1)}{2}} - 1}{11}\right)$$

$$= (12)^{\frac{(q-1)}{2}}(4) + \left(\frac{1}{11}\right)(12)^{\frac{(q-1)}{2}} - \left(\frac{1}{11}\right)$$

2

$$\sum_{k=0}^{n}((12)^k) = \frac{(12)^{n+1} - 1}{11} \quad \langle n \leftarrow \left\lfloor \left(\frac{n}{2}\right) \right\rfloor \rangle$$

$$\sum_{k=0}^{\left\lfloor \left(\frac{n}{2}\right) \right\rfloor}((12)^k) = \left(\frac{(12)^{\left(\left\lfloor \left(\frac{n}{2}\right) \right\rfloor + 1\right)} - 1}{11}\right)$$

3

$$\sum_{k=0}^{n}((12)^k) = \frac{(12)^{n+1} - 1}{11} \quad \langle n \leftarrow \left\lfloor \left(\frac{(n-1)}{2}\right) \right\rfloor \rangle$$

$$\sum_{k=0}^{\left\lfloor \left(\frac{(n-1)}{2}\right) \right\rfloor}(12)^k = \left(\frac{(12)^{\left(\left\lfloor \left(\frac{(n-1)}{2}\right) \right\rfloor + 1\right)} - 1}{11}\right)$$

4

$$\sum_{k=0}^{n}((12)^k)$$

$$= Sm((12)^k, n)$$

$$= \frac{1 - (12)^{(n+1)}}{1 - (12)}$$

$$= \frac{(12)^{(n+1)} - 1}{11}$$

5

$$\varphi(0) = (12)^{\frac{0}{2}}(3) + \left(\frac{1}{11}\right)(12)^{\frac{0}{2}} - \left(\frac{1}{11}\right)$$

$$\varphi(1) = (12)^{\frac{(1-1)}{2}}(4) + \left(\frac{1}{11}\right)(12)^{\frac{(1-1)}{2}} - \left(\frac{1}{11}\right)$$

$$\varphi(2) = (12)^{\frac{2}{2}}(3) + \left(\frac{1}{11}\right)(12)^{\frac{2}{2}} - \left(\frac{1}{11}\right)$$

$$\varphi(3) = (12)^{\frac{(3-1)}{2}}(4) + \left(\frac{1}{11}\right)(12)^{\frac{(3-1)}{2}} - \left(\frac{1}{11}\right)$$

$$\varphi(4) = (12)^{\frac{4}{2}}(3) + \left(\frac{1}{11}\right)(12)^{\frac{4}{2}} - \left(\frac{1}{11}\right)$$

$$Sm(\varphi, 4) = (3)\left((12)^0 + (12)^1 + (12)^2\right) + (4)\left((12)^0 + (12)^1\right)$$
$$+ \left(\frac{1}{11}\right)\left(\left((12)^0 + (12)^1 + (12)^2\right) + \left((12)^0 + (12)^1\right)\right)$$
$$- \left(\frac{1}{11}\right)(4+1)$$

$$Sm(\varphi, n) = \left((3)\sum_{k=0}^{\left\lfloor \left(\frac{n}{2}\right) \right\rfloor}(12)^k + (4)\sum_{k=0}^{\left\lfloor \left(\frac{(n-1)}{2}\right) \right\rfloor}(12)^k\right) + \left(\frac{1}{11}\right)\left(\sum_{k=0}^{\left\lfloor \left(\frac{n}{2}\right) \right\rfloor}(12)^k + \sum_{k=0}^{\left\lfloor \left(\frac{(n-1)}{2}\right) \right\rfloor}(12)^k\right) - \left(\frac{1}{11}\right)(n+1)$$

$$= \left((3) + \left(\frac{1}{11}\right)\right)\sum_{k=0}^{\left\lfloor \left(\frac{n}{2}\right) \right\rfloor}(12)^k + \left((4) + \left(\frac{1}{11}\right)\right)\sum_{k=0}^{\left\lfloor \left(\frac{(n-1)}{2}\right) \right\rfloor}(12)^k - \left(\frac{1}{11}\right)(n+1)$$

$$= \left(\frac{34}{11}\right)\sum_{k=0}^{\left\lfloor \left(\frac{n}{2}\right) \right\rfloor}(12)^k + \left(\frac{45}{11}\right)\sum_{k=0}^{\left\lfloor \left(\frac{(n-1)}{2}\right) \right\rfloor}(12)^k - \left(\frac{1}{11}\right)(n+1)$$

$$= \left(\frac{34}{11}\right)\left(\frac{(12)^{\left(\left\lfloor \left(\frac{n}{2}\right) \right\rfloor + 1\right)} - 1}{11}\right) + \left(\frac{45}{11}\right)\left(\frac{(12)^{\left(\left\lfloor \left(\frac{(n-1)}{2}\right) \right\rfloor + 1\right)} - 1}{11}\right) - \left(\frac{1}{11}\right)(n+1)$$

6 (3a) Consider the following recurrence relation for $\varphi(n)$ and **answer the questions**:

$$\begin{cases} \varphi(0) \coloneqq c \\ \forall n \in \mathbb{N} \quad \varphi(n+1) \coloneqq \big(\varphi(n)\big)a + b \end{cases}$$

Show calculations on the facing side.

Rewrite the recursion using your values of $a, b,$ and c.

$$\begin{cases} \varphi(0) \coloneqq (2) \\ \forall n \in \mathbb{N} \quad \varphi(n+1) \coloneqq \big(\varphi(n)\big)(\ |\ ) + (4) \end{cases}$$

Show computations on the facing side for making a conjecture:

2 (i) Conjecture:

$$\forall n \in \mathbb{N} \quad \varphi(n) = \quad 2 \ + \ (4)(n)$$

4 (i) Prove your conjecture by induction:

3a.i.0

$$\frac{\begin{array}{rcl}\varphi(n+1) &=& (\varphi(n))(1)+4 \\ \varphi(0+1) &=& (\varphi(0))(1)+4 \\ &=& (2+0+0)(1)+4\end{array}}{\varphi(1) = 2+4} \langle n \leftarrow 0 \rangle$$

3a.i.1

$$\frac{\begin{array}{rcl}\varphi(n+1) &=& (\varphi(n))(1)+4 \\ \varphi(1+1) &=& (\varphi(1))(1)+4 \\ &=& (2+4+0)(1)+4\end{array}}{\varphi(2) = 2+4+4} \langle n \leftarrow 1 \rangle$$

3a.i.2

$$\frac{\begin{array}{rcl}\varphi(n+1) &=& (\varphi(n))(1)+4 \\ \varphi(2+1) &=& (\varphi(2))(1)+4 \\ &=& (2+4+4)(1)+4\end{array}}{\varphi(3) = 2+4+4+4} \langle n \leftarrow 2 \rangle$$

3a.i.3

$$\begin{array}{rcl}\varphi(0) &=& 2 \\ \varphi(1) &=& 2+4 \\ \varphi(2) &=& 2+4+4 \\ \varphi(3) &=& 2+4+4+4\end{array}$$

$$\varphi(n) = 2+(4)(n)$$

3a.ii.0

$$\forall n \in \mathbb{N}\Big(P(n) :\equiv \big(\varphi(n) = 2+(4)(n)\big)\Big)$$

3a.ii.1

Bc

LS	RS
$= \varphi(0)$	$= 2+(4)(0)$
$= 2$	$= 2+0$
	$= 2$

$LS = 2 = RS$

Bc

3a.ii.2

Rcs

$P(0)\ P(1)\ \dots\ P(n)$

$$\begin{array}{rcl} & & LS \\ &=& \varphi(n+1) \\ &=& ((\varphi(n))(1)+4 \\ &=& (2+(4)(n))(1)+4 \\ &=& 2+(4)(n)+4 \\ &=& 2+(4)(n+1) \\ &=& RS\end{array}$$

$P(n+1)$

RcS

3a.ii.3

$$\frac{\dfrac{\forall n \in \mathbb{N}\left(P(0)\ \left(\frac{P(0)\ P(1)\ \dots\ P(n)}{P(n+1)}\right)\right)}{\forall n \in \mathbb{N}(P(n))}}{\forall n \in \mathbb{N}\big(\varphi(n) = 2+(4)(n)\big)}$$

6 (3b) Prove, following the method shown in class, that: $(a+b+1)^{\frac{1}{a+b+2}}$ **is irrational**, that is: $\textit{Rtl}\left((a+b+1)^{\frac{1}{a+b+2}}\right)$.

First rewrite the problem using your values of a and b.

My Problem:

$$\textit{Rtl}\left(((\ ) + (\ ) + 1)^{\frac{1}{(\ )+(\ )+2}}\right)$$

that is: $\textit{Rtl}\left((\ )^{\frac{1}{(\ )}}\right)$

3b.i

$$\neg\left(\cancel{Rtt}\left((5)^{\frac{1}{(7)}}\right)\right)$$

$$Rtl\left((5)^{\frac{1}{(7)}}\right)$$

$$\exists!\, p,q \in \mathbb{N}\left(\left(gcd(p,q)=1\right) \wedge \left((5)^{\frac{1}{(7)}} = \frac{p}{q}\right)\right)$$

$$\left(gcd(p,q)=1\right) \quad \left((5)^{\frac{1}{(7)}} = \frac{p}{q}\right)$$

$$(5)^{\frac{1}{(7)}} = \frac{p}{q}$$

$$\left((5)^{\frac{1}{(7)}}\right)^7 = \left(\frac{p}{q}\right)^7$$

$$5 = \frac{p^7}{q^7}$$

$$5q^7 = p^7$$

$$prm(5) \quad 5 \mid p^7$$

$$5 \mid p$$

$$\exists!\, r \in \mathbb{N}\,(p = 5r)$$

$$5q^7 = (5r)^7$$

$$5q^7 = 5^7 r^7$$

$$q^7 = 5^6 r^7$$

$$(5)5^5 r^7 = q^7$$

$$prm(5) \quad 5 \mid q^7$$

$$5 \mid q$$

$$5 \mid p \quad 5 \mid q$$

$$5 \mid (gcd(p,q))$$

$$5 \mid (gcd(p,q)) \qquad gcd(p,q) = 1$$

$$5 \mid 1$$

$$\perp$$

$$\neg\left(\neg\left(\cancel{Rtt}\left((5)^{\frac{1}{(7)}}\right)\right)\right)$$

$$\cancel{Rtt}\left((5)^{\frac{1}{(7)}}\right)$$